ROB RANDHAWA

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SUMMARY

Senior Big Data Engineer with over 11 years of experience designing, building, and operating large-scale data pipelines across cloud and on-premise environments. Track record of owning pipelines tied to $100M+ in annual revenue, leading cross-functional delivery across 5+ teams, and driving measurable cost reduction ($1M+ in storage savings). Deep expertise in the AWS ecosystem, Databricks, Spark, and Snowflake, with a recent focus on AI-assisted development and LLM-powered internal tooling.

TECHNICAL SKILLS

Languages: Python, Java, Scala, Bash, SQL

Big Data & Cloud: Spark, Databricks, HDFS, MapReduce, Kafka, AWS (EMR, S3, Lambda, SQS, SNS, DynamoDB, Glue, Sagemaker, EKS, ECR, ECS, CloudFormation), GCP (BigQuery, GCS), Airflow, Oozie, NiFi

Databases: Snowflake, PostgreSQL, MySQL, DynamoDB, Hive, Impala, HBase, Oracle, IBM DB2

DevOps & Infra: Docker, Kubernetes, Terraform, Helm, Jenkins, Ansible, CloudWatch, PagerDuty

Frameworks & Tools: FastAPI, Flask, Pandas, R, Cursor, Claude, ChatGPT (AI-assisted development), LangChain/LangGraph, dbt, Prometheus

Source Control: Git, Mercurial

PROFESSIONAL EXPERIENCE

Senior Big Data Engineer | Integral Ad Science

Aug 2021 – Mar 2026

Led data pipeline strategy for Walled Garden integrations — pulling ad measurement data (viewability, brand safety) from major platforms and transforming it into internal data models consumed by client-facing dashboards and reports.

- Built and maintained pipelines generating over $100M in annual revenue; coordinated delivery across 5 cross-functional teams (Product, Data Engineering, Data Science, Customer Excellence, Cloud Reliability Engineering); mentored junior engineers
- Designed and built new ingestion pipelines for Twitter (video/display viewability and brand safety), Mail.ru (viewability), YouTube (GARM brand safety framework), and Meta/Facebook platforms (brand suitability impression data for GARM scoring across Facebook, Instagram, and Audience Network)
- Automated and enhanced Meta viewability audit review processes (display and video) for Media Rating Council (MRC) accreditation; updated formulations, metrics, and documentation
- Created a Databricks table monitoring process to detect and enforce TTL (Time-to-Live) properties — saving over $1M in storage costs
- Built a FastAPI application backed by an LLM to answer internal team questions about Walled Gardens data table schemas and data delay statuses; deployed to AWS EKS via Jenkins pipelines and Helm charts
- Maintained legacy HDFS systems while migrating pipelines to cloud providers

Stack: Python 3, Scala, Databricks, Airflow, Snowflake, AWS (CloudFormation, CloudWatch, IAM, SNS, SQS, S3, EMR, DynamoDB, ECR, ECS), Jenkins, Docker, Terraform, Helm, Kubernetes, GCP (BigQuery, GCS), PagerDuty, dbt, Cursor

Senior Big Data Engineer / Big Data Engineer | Fidelity Investments

Feb 2018 – Aug 2021

Progressed from Big Data Engineer to Senior across two major platform initiatives: building a greenfield data lake and architecting a new ML model deployment platform.

Machine Learning Platform

Sep 2019 – Aug 2021

- Migrated production ML models from Greenplum to AWS SageMaker using both Bring Your Own Container (BYOC) and Bring Your Own Algorithm (BYOA) patterns
- Designed and published migration standards adopted by other development teams
- Architected a schema to describe and govern model deployment; stood up a PostgreSQL RDS Aurora Serverless cluster (VPC-isolated) to house model registry data
- Built an event-driven model invocation flow using S3 events, manifest files, SNS, SQS, and Lambda; created a parallel flow for auto-deploying trained models upon metadata file arrival
- Mentored junior developers and created onboarding documentation for the platform

Stack: AWS (SageMaker, Lambda, CloudFormation, Step Functions, RDS, S3, ECR, CloudWatch, SNS, SQS), Python 3, PostgreSQL, Docker, Jenkins, Flask, Gunicorn, QuickSight, Elasticsearch, Kibana

PI Data Lake

Feb 2018 – Aug 2019

- Built an event-driven, serverless data lake for Fidelity's Personal Investing division to reduce data bifurcation and enable data scientist schema discovery
- Deployed and administered an on-premise NiFi cluster with custom Processors to optimize memory and content repository usage for S3 uploads
- Wrote Java/Scala Spark jobs (EMR and Glue ETL) to flatten nested structures and output Parquet/Snappy files; templated ingestion framework with CloudFormation including dynamic PII-aware S3 bucket policies
- Centralized logging to Elasticsearch/Kibana; authored Jenkins DSL Groovy pipelines for automated continuous delivery

Stack: AWS (Lambda, CloudFormation, Step Functions, EMR, Glue, S3, Athena, SNS, SQS, Batch, ECR), Java 8, Scala, Python, NiFi, Zookeeper, Docker, Ansible, Jenkins, Elasticsearch, Kibana

Software Engineer | IBM Watson Health *Nov 2016 – Jan 2018*

- Implemented RESTful file-sharing service (folder/file privileges) for a client-facing platform using Spring 4 and Java 8; wrote controller and service layers integrating with file share server APIs
- Enhanced a distributed data processing application: optimized Spark job performance (checkpointing, Kryo serialization, SQL plan truncation), wrote Spark UDFs for complex business logic, and updated Ansible deployment scripts and Liquibase migrations
- Researched PMML and PFA language specifications for a dynamic event grouping POC; evaluated engine licensing compliance
- Migrated a legacy Truven Healthcare data lake onboarding project from Mercurial to Git; increased test coverage and automated schema version retrieval from Artifactory

Hadoop Developer | Ana-Data Consulting, Inc. *Jul 2014 – Aug 2016*

- Executed TPC-H benchmark testing comparing Hive, Spark SQL, Hawq, and VectorH on datasets from 1GB to 500GB; automated query submission with Python/shell scripts and visualized results in Tableau
- Extended Hive capabilities with UDFs and custom SerDes; optimized query performance with static/dynamic partitioning and bucketing; automated ETL scheduling with Oozie coordinators
- Migrated commercial telemarketing analytics from SAS to a Hadoop MapReduce/Hive framework, reducing lead time from 5 days to 4 hours

GITHUB PROJECTS

data_processing *github.com/rubinder/data_processing* — Python, PySpark, Shell, AWS, Databricks, dbt, Flink, Airflow, FastAPI, Apache Iceberg
Repository demonstrating a multi-platform (local, Airflow, AWS, Databricks) deployment of data processing applications that process simulated data generated by the accompanying web server data.

wiz_scheduler *github.com/rubinder/wiz_scheduler* — Python, FastAPI, LangGraph, Next.js, PostgreSQL, Supabase, Claude API, Prometheus, Docker, Vercel
AI-powered multi-tenant employee scheduling platform built for commercial release. Key engineering highlights: LangGraph agentic pipeline with defensive LLM output validation, constraint-based scheduling across multiple locations with cross-location double-booking prevention, usage-based billing engine with Stripe integration and block-based overage tracking, Prometheus metrics with multiprocess worker support including request latency histograms and DB pool gauges, bidirectional 7shifts and Deputy integrations with idempotent external-ID-based reconciliation, JWT-based multi-tenant isolation with ownership group hierarchy, GDPR data export and consent tracking, and 19-language i18n with RTL support. Available at wizscheduler.com

logistics_agents *github.com/rubinder/logistics_agents* — Python, Anthropic API, FastAPI, React, TypeScript, Postgres, Kafka (Redpanda), Docker, AWS, Terraform
Multi-agent logistics decisioning system built to demonstrate agent quality as an engineering discipline rather than prompt tinkering. A fixed-DAG pipeline (orchestrator → three specialist agents → synthesis) turns an inbound shipment notification into a validated, structured Decision, with every node independently gradable. The differentiator is the eval and observability layer wrapped around it: a labeled eval dataset, a hybrid composite grader (deterministic checks + LLM-as-judge), and a record/replay fixture cache so the regression suite runs deterministically and free on every commit, while nightly live runs compare quality, latency, and cost across Opus, Sonnet, and Haiku. Each node emits a trace record (I/O, latency, tokens, cost, model) persisted to Postgres and streamed over Kafka; a publicly deployed React dashboard renders the DAG live over SSE, replays stored traces step-by-step, and charts quality-over-time, with rate-limited visitor-triggered runs guarded by a hard budget-cap ledger. Available at dh95veq0vfppo.cloudfront.net

Financial Document RAG Pipeline *github.com/rubinder/financial_doc_rag_pipeline* — Python, FastAPI, Qdrant, Prefect, Langfuse, DVC
Designed and built a Prefect-orchestrated ingest pipeline that pulls SEC 10-K filings from EDGAR, parses unstructured HTML into section-aware chunks, embeds with OpenAI, and upserts into Qdrant delivering production-ready vector data to a RAG application with retries, typed task boundaries, and reproducible flow runs. Engineered the data schema for LLM retrieval: section-aware payloads (item_id, item_title, ticker, corpus_version) enable filtered semantic search, and a two-stage retrieve→rerank path (cosine top-K × 4 → ms-marco cross-encoder, sigmoid-calibrated to a [0,1] confidence score) materially improved answer precision on cross-section queries. Implemented end-to-end data lineage and observability across the AI inference path: every /query emits Langfuse spans for embed/search/rerank/synth, a SQLite-backed /metrics endpoint exposes retrieval-quality KPIs (mean cosine, low-confidence rate, latency p50/p95, top-retrieved sections), and a corpus_version SHA join key links each trace to its DVC-tracked raw filings making any production answer fully replayable against its exact source corpus.

CERTIFICATIONS

- AI Agents in LangGraph — DeepLearning.AI (Mar 2026)
- AI Engineering Specialization — ByteByteGo (Nov 2025)
- CKAD: Certified Kubernetes Application Developer — Linux Foundation (May 2022, exp. May 2025)
- AWS Certified Developer (Oct 2018, exp. Oct 2021)
- Hortonworks Data Platform Certified Developer (Feb 2016)

EDUCATION

Bachelor of Arts, Economics — University of Miami, 2014

Interests: Basketball, Surfing, Cooking, Capoeira, Weight Lifting, Yoga